

SAURABH TIWARI

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PROFESSIONAL SUMMARY

Motivated Computer Science student at PSIT Kanpur with a strong foundation in Java, Data Structures & Algorithms, and the MERN stack. Passionate about building scalable web applications and AI-based solutions that solve real-world problems. Experienced in developing full-stack platforms with modern technologies and committed to writing clean, efficient, and maintainable code. Eager to contribute to innovative engineering teams and grow as a software developer.

EDUCATION

Pranveer Singh Institute of Technology (PSIT), Kanpur

8.42 CGPA (1st semester)

B.Tech in Computer Science | 1st Division | Currently Pursuing

Kendriya Vidyalaya

2024

Senior Secondary Education (PCM)

TECHNICAL SKILLS

Languages: Java, JavaScript (ES6+), HTML5, CSS3

Core Concepts: Data Structures & Algorithms (DSA), Object-Oriented Programming (OOP)

Frameworks & Libraries: React.js, Node.js, Express.js

Database & Tools: MongoDB, Git, GitHub, REST APIs, WebSocket, WebRTC

PROJECTS

Blood Connect (*MERN Stack | Full Stack Project*) – Web Application

- Developed a full-stack MERN (MongoDB, Express.js, React.js, Node.js) platform to connect blood donors with recipients during emergency situations.
- Implemented secure user authentication and authorization using JWT, ensuring safe and reliable access control.
- Built real-time donor search functionality with location-based filtering to quickly find nearby donors.
- Designed and developed scalable RESTful APIs for efficient data handling and seamless communication between frontend and backend.
- Created a responsive and user-friendly React interface to enhance user experience across devices.
- Optimized performance and database queries to ensure faster response time and smooth operation under load.
- Focused on real-world usability, enabling quick registration, search, and communication between donors and recipients.

AI Interview Platform (*React, Node.js | Full Stack Project*)

- Developed an AI-powered interview preparation platform simulating real-world technical interview rounds.
- Designed a multi-stage interview system including aptitude, DSA, and coding rounds for structured practice.
- Integrated DSA-based coding questions with redirection to coding platforms (like LeetCode-style environment).
- Implemented dynamic question rendering and evaluation logic to enhance user engagement.
- Built a responsive frontend using React and managed backend operations with Node.js.
- Focused on scalability and modular architecture to support future AI-based evaluation features.
- Aimed to help students improve problem-solving skills and prepare for product-based companies.

Real-Time Communication App (*React, Node.js, WebSocket, WebRTC | Full Stack Project*)

- Built a real-time communication platform enabling video calls, audio calls, and instant messaging, similar in spirit to Google Meet.
- Implemented a WebSocket-based signaling server to establish peer connections and enable low-latency, real-time communication.
- Integrated WebRTC for peer-to-peer audio and video streaming with minimal latency and buffering.
- Developed a collaborative real-time whiteboard feature, allowing multiple participants to draw, annotate, and brainstorm together during a call.
- Designed multi-user meeting rooms supporting simultaneous video conferencing sessions with screen sharing.

Tech Stack: React.js, Node.js, Express.js, Socket.io/WebSocket, WebRTC, MongoDB

Sign Language Translator (*AI/ML + Web | Computer Vision Project*)

- Developed an AI-based web application to translate sign language gestures into text and speech.
- Implemented real-time hand gesture recognition using computer vision techniques.
- Integrated camera-based input to detect and classify hand signs efficiently.
- Built features like sign-to-text, text-to-speech, and voice-to-sign for complete communication flow.
- Designed the system to assist specially-abled users in overcoming communication barriers.
- Optimized model performance for faster and more accurate gesture detection.
- Combined AI/ML with web technologies to deliver a practical and impactful solution.

CERTIFICATIONS

Meta PyTorch OpenEnv Hackathon – Round 1 Qualifier *in collaboration with Scaler School of Technology on April 26, 2026*

- Successfully qualified Round 1 of the Meta PyTorch OpenEnv Hackathon as part of a two-member team, alongside teammate Shubhi Shukla.
- Applied AI/ML concepts, problem-solving, and real-world innovation to build a working solution under the hackathon's evaluation criteria.
- Advancing to the next round of the hackathon.

